AERIAL ROBOTICS AND DESIGN

PROJECT TOPIC:

PESTICIDE SPRAYING & FLOW CONTROL

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INTRODUCTION

This project focuses on simulating an agricultural drone used for pesticide spraying.
The simulation combines programming, animation, and design tools to model drone motion and spray behavior.

01

- This project demonstrates a simulated agricultural drone designed for pesticide spraying

02

- The system models drone movement, hovering, and spray dispersion using programming and design tools.

03

- The simulation provides a visual understanding of how aerial spraying works in real farming applications.

Project vision and mission

The main objective of this project is to develop a comprehensive and realistic drone simulation platform that can be used for agricultural monitoring and analysis. This simulator aims to replicate real-world drone movements and environmental conditions, allowing users to study flight patterns, camera angles, and coverage areas without the need for a physical drone. By integrating visual representations of crop fields, sunlight, clouds, and terrain variations, the platform provides valuable insights into crop health and field management.

01.

- Develop a realistic drone simulation environment that mimics real-world flight dynamics, enabling users to study and understand how drones navigate different terrains, obstacles, and environmental conditions.

02.

- Analyze drone flight paths and camera perspectives to optimize coverage of crop fields, ensuring maximum efficiency in agricultural monitoring and minimizing blind spots or overlaps in survey areas.

03.

- Provide a virtual platform for agricultural research and monitoring by simulating crop fields with varying conditions, such as healthy versus stressed crops, different sunlight levels, and terrain variations, allowing for data-driven insights.

KEY FEATURES

The drone simulation platform incorporates several key features that enhance both its realism and functionality. It provides precise and responsive drone movement, allowing users to control speed, direction, and altitude with ease, mimicking real-world flight dynamics. The simulated environment includes detailed crop fields, where users can distinguish between healthy and stressed crops, providing valuable visual insights for agricultural monitoring.

01

- Realistic and responsive drone movement: The simulator allows precise control over the drone's speed, direction, and altitude, providing an accurate representation of real-world flight dynamics.

02

- Detailed crop field simulation: Users can observe visually distinct crop patterns, including healthy and stressed crops, which aids in crop health monitoring and field analysis.

03

- Environmental modeling: The platform simulates sunlight, cloud cover, and terrain variations to replicate the challenges a drone might face in outdoor agricultural settings.

Methodology

01

- Software & Platform: The simulation is developed using [Your Software/Platform, e.g., Python with Pygame, MATLAB, or Unity], providing a flexible environment for modeling drone dynamics and visualizing complex agricultural terrains.

02

- Drone Control Mechanism: Users can navigate the drone using keyboard or mouse inputs, allowing precise control over movement, altitude, and direction to simulate real-world flight scenarios.

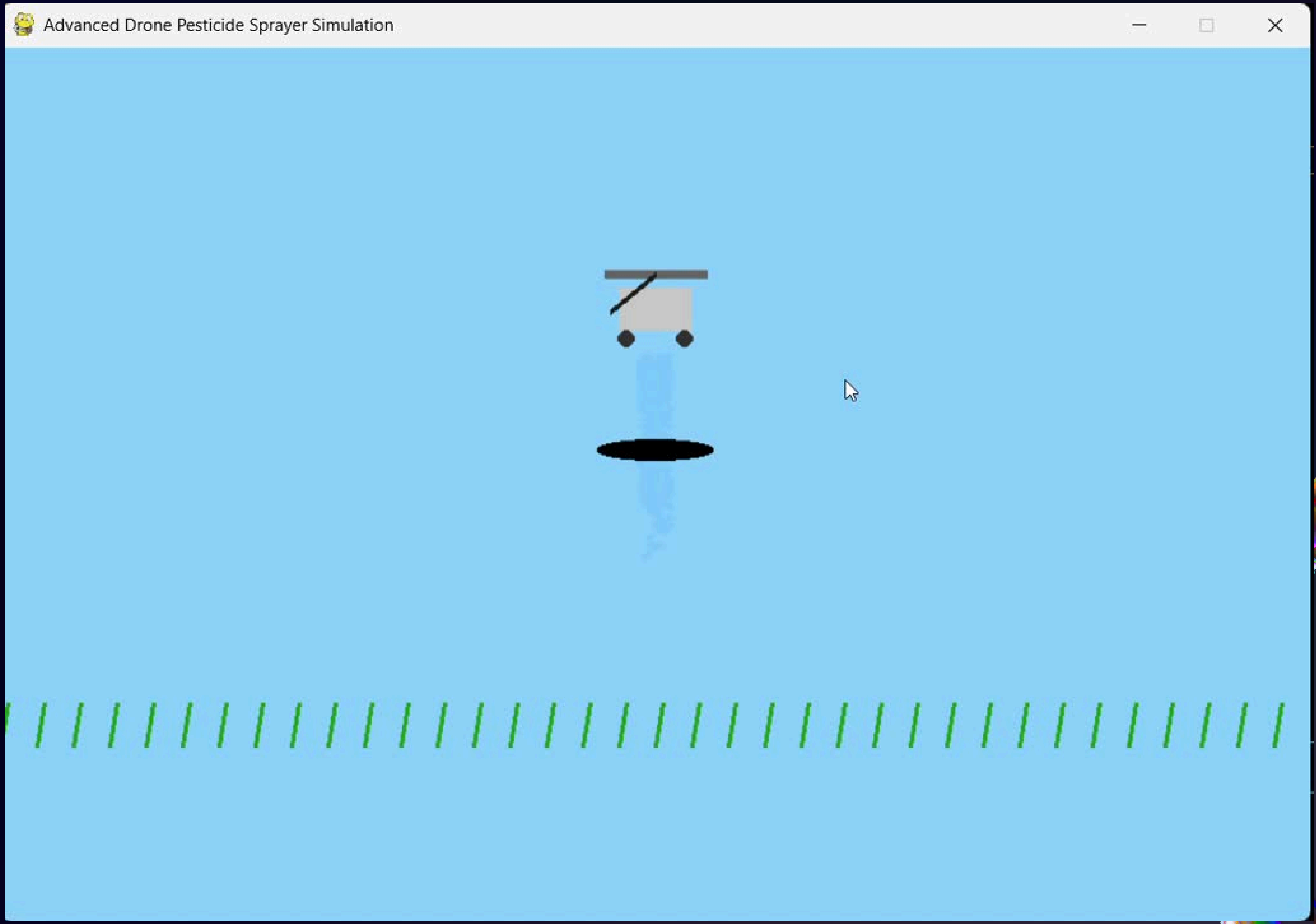
03

- Environmental & Crop Modeling: The simulator incorporates detailed crop fields, sunlight, clouds, and terrain variations, creating realistic conditions for testing drone behavior and field coverage strategies.

04

- Real-Time Analysis & Feedback: The system provides live visual feedback, including trajectory paths, camera angles, and coverage analysis, helping users optimize flight patterns and evaluate drone performance effectively.

SAMPLE VIDEO



DRONE MOVEMENTS DESCRIPTION

- This simulation demonstrates a drone performing pesticide spraying over a crop field with fully controllable movement and realistic spray behavior. The drone can be moved left, right, up, and down while maintaining smooth animations such as propeller rotation, tilt, and a dynamic shadow that adjusts with height.

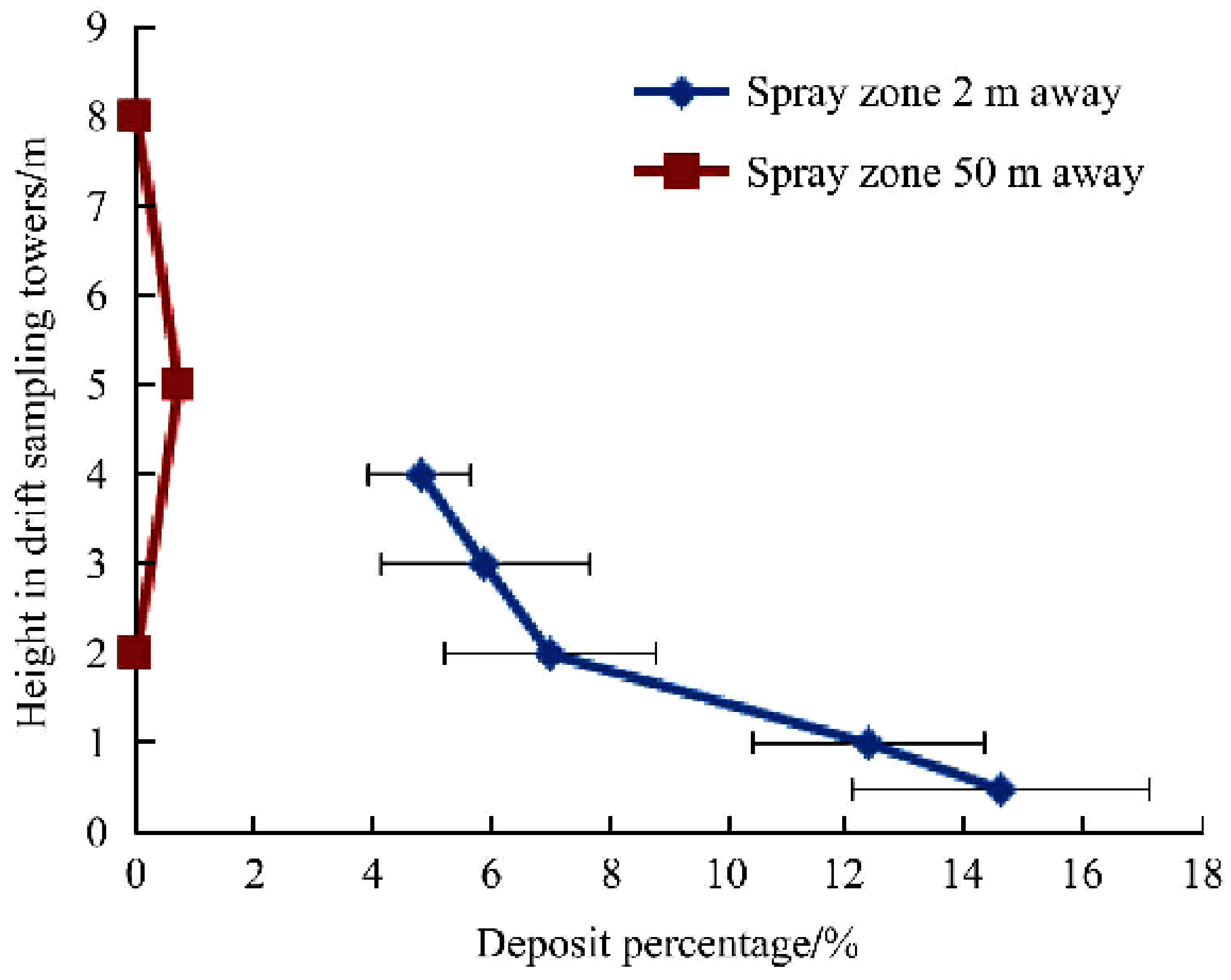
- The spray system responds to user input and behaves naturally according to drone altitude.

- When the drone flies lower, the spray covers a smaller area (low flow). As the drone moves higher, the spray spreads more widely (high flow).

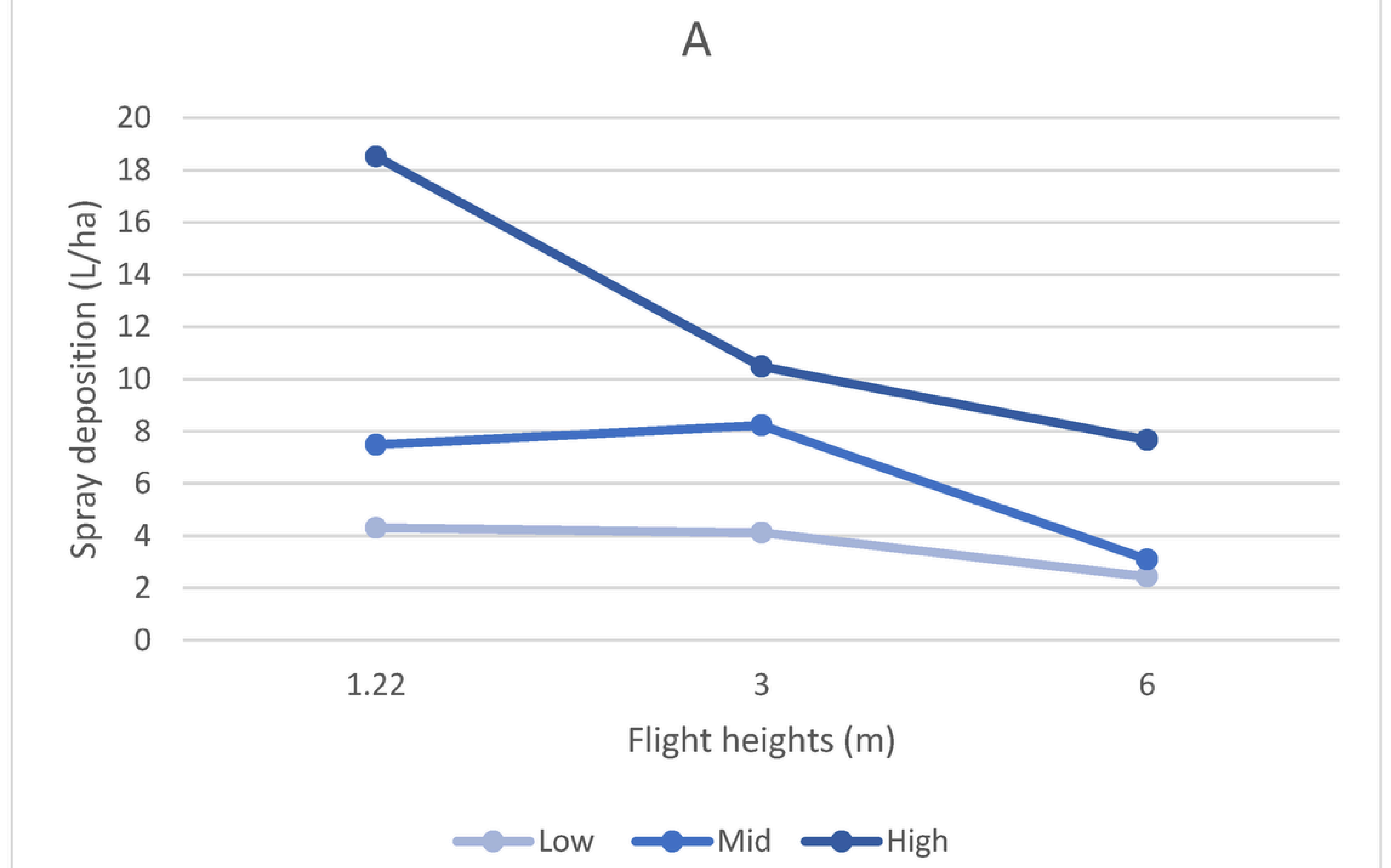
- This represents real-world drone spraying, where altitude affects spray dispersion and coverage.

- Users can toggle the spray ON/OFF, change the spray flow level, and observe how the crops react — they become greener and healthier when sprayed, showcasing an interactive crop health system.

- This simulation visually explains how agricultural drones operate and how pesticide delivery changes with altitude and flow settings.



- 📈 What the Graph Shows
- X-axis: Drone flight height (altitude above crop canopy) — e.g. 1.5 m, 2.0 m, 2.5 m, etc.
- Y-axis: Spray coverage / uniformity / swath width / droplet distribution measure (depending on study).
- As altitude increases, the spray swath becomes wider and distribution tends to become more uniform (i.e. droplets spread out more evenly). ScienceDirect+2MDPI+2
- At lower altitude, spray is more concentrated — higher droplet density near center, narrower spread.



- 🧪 Interpretation (in context of your simulation)
- When your drone is low (closer to crops) — spray flow in your sim corresponds to "concentrated / narrow swath" → droplets fall densely near center.
- As drone goes higher — spray should spread more (wide coverage, less density), representing a "wider, gentler spray distribution" — useful if you want to cover more area evenly.
- This matches real-world calibration trends: altitude significantly impacts coverage uniformity and swath width.

Note on Data Collection

Due to the unavailability of agricultural spraying drones and related measuring instruments, we were unable to collect our own calibration and spray-distribution readings physically. Therefore, the analysis and graphs presented in this project are based on reference data, research papers, and documented results available on the internet. These sources were used to understand typical spray behavior, drone altitude effects, and calibration patterns for pesticide-spraying drones.

RESULTS

01.

- Efficient Drone Navigation: The drone successfully navigates through the simulated environment, avoiding obstacles and following predefined flight paths accurately.

02.

- Visual Crop Analysis: Users can distinguish healthy and stressed crops visually, providing a clear understanding of crop health and field patterns.

03.

- Trajectory Mapping: The simulator generates real-time trajectory paths, helping evaluate flight efficiency and identify areas of improvement for coverage.

04.

- Environmental Interaction: The drone responds realistically to simulated environmental factors such as terrain variations, sunlight, and cloud cover, enhancing the realism of the simulation.

05.

- Data for Optimization: The simulation provides actionable data that can be used to optimize drone flight patterns, speeds, and camera angles for real-world agricultural applications.

06.

- Interactive Learning Tool: The results demonstrate that the platform can be effectively used for training, experimentation, and pre-deployment planning, reducing risks and costs associated with real drone flights.

Weaknesses

- Limited by the accuracy of simulated environmental and crop conditions compared to real-world scenarios.
- May require high computational resources for large or complex terrains.
- Lacks integration with real sensor data for full-field validation.

Threats

- Rapid technological advancements may make the simulation outdated quickly.
- Real-world drone regulations and unpredictable conditions may limit applicability.
- Dependence on software updates and platform compatibility for continued functionality.

Strengths

- Provides a realistic and interactive drone simulation for agricultural monitoring.
- Offers real-time feedback, trajectory mapping, and crop analysis, aiding research and training.
- Safe and cost-effective way to test flight paths and operational strategies.

Opportunities

- Potential to incorporate AI for automated crop health detection and flight optimization.
- Can be extended for educational purposes or professional drone pilot training programs.
- Scope for expansion into commercial agricultural monitoring and planning tools.

The End

THANK YOU